SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering

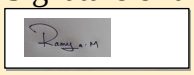
ASSESSMENT TOOL 3 – OUTPUT PREDICTION

Course Code	**CSA0315**	**Course Title**	**Data Structures**
CO Assessed	**CO1**	**Assessment**	**Assessment Tool 3: Output Prediction**
Weightage	**30% of CIA**	**Total Marks**	**100**
CO1 Statement	**Apply suitable data structures and ADTs for efficient problem solving by analyzing timeand space complexity. (BL3, BL4)**		
Student Name	**Ramya M**	**Reg. No.**	**1921260148**
Section/Batch	**Slot D / 2026**	**Date**	**10-10-2026**

Code of Conduct:

I, Ramya M (1921260148), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

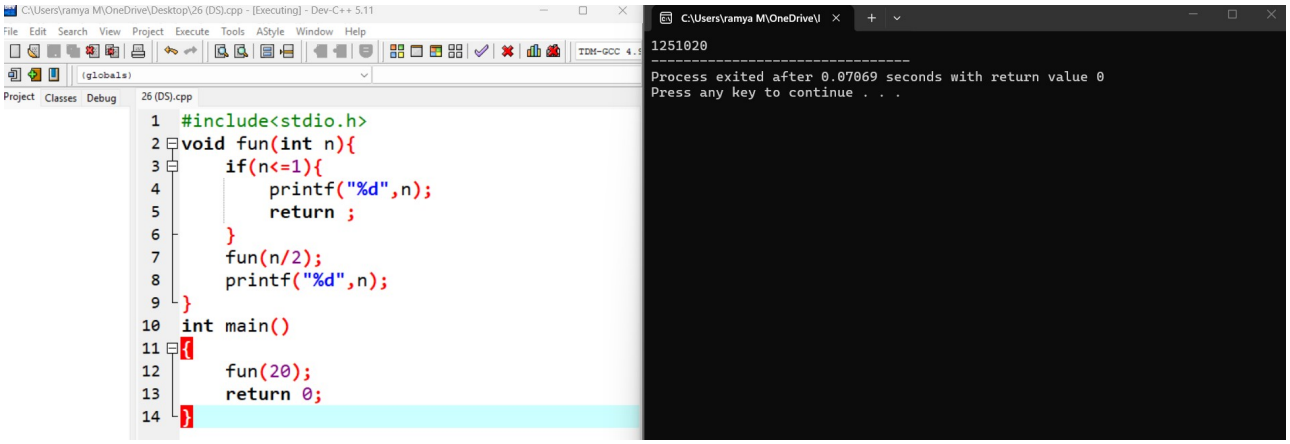
Signature of the student:



Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Output Accuracy	30%	Predicts all program outputs correctly, including exact values and output sequence.	Predicts most outputs correctly with minor errors.	Predicts some outputs correctly but makes significant errors.	Fails to predict most program outputs correctly.	
Conceptual Understanding	25%	Demonstrates excellent understanding of ADT, arrays, recursion, data structures, and C programming concepts.	Demonstrates good understanding with minor conceptual errors.	Demonstrates partial understanding with several conceptual errors.	Demonstrates inadequate understanding of fundamental concepts.	
Logical Tracing and Execution Flow	20%	Traces loops, recursive calls, array updates, and function execution accurately and systematically.	Traces execution correctly with a few minor mistakes.	Shows incomplete tracing with errors in execution flow.	Unable to trace program execution logically.	
Time and Space Complexity Analysis	15%	Correctly identifies and justifies best-, average-, and worst-case complexity and auxiliary space requirements.	Identifies most complexities correctly with minor errors.	Identifies some complexities but provides incomplete or incorrect reasoning.	Unable to identify or explain algorithm complexity.	
C Syntax and Runtime Behavior	10%	Correctly interprets C expressions, variable updates, function calls, and undefined behavior.	Understands most C execution rules with minor mistakes.	Shows limited understanding of execution rules and expression evaluation.	Misinterprets fundamental C language rules and program behavior.	

LAB PROGRAMS

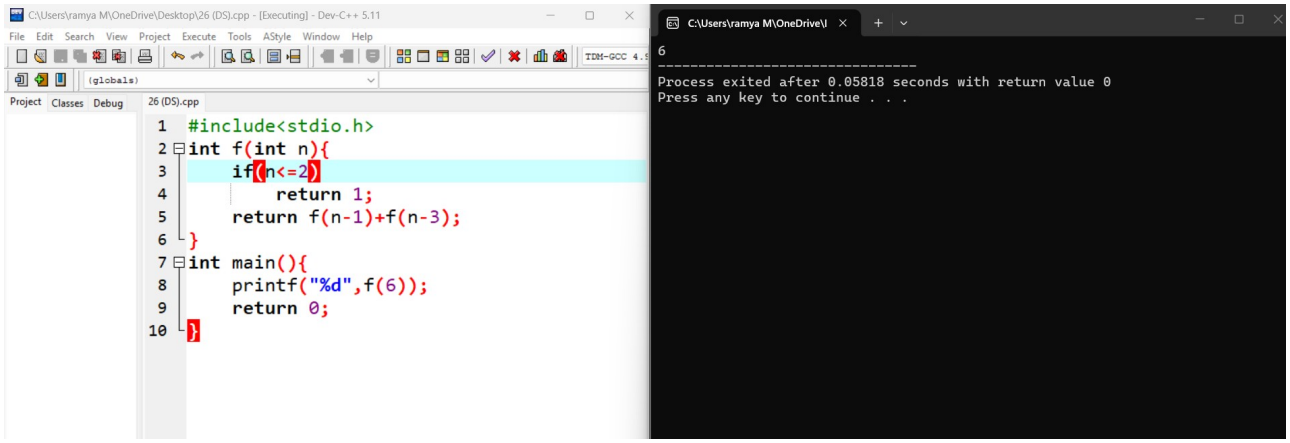
Question 26. Recursive printing with multiplication



```
#include<stdio.h>
void fun(int n){
    if(n<=1){
        printf("%d",n);
        return ;
    }
    fun(n/2);
    printf("%d",n);
}
int main()
{
    fun(20);
    return 0;
}
```

```
1251020
--------------------------------
Process exited after 0.07069 seconds with return value 0
Press any key to continue . . .
```

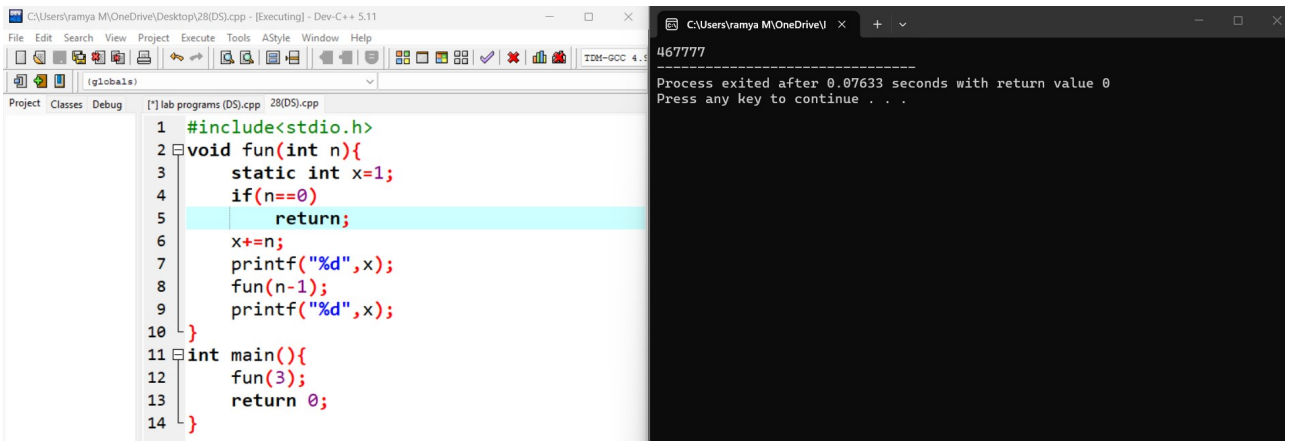
Question 27. Recursion and multiple recursive calls



```
#include<stdio.h>
int f(int n){
    if(n<=2)
        return 1;
    return f(n-1)+f(n-3);
}
int main(){
    printf("%d",f(6));
    return 0;
}
```

```
6
--------------------------------
Process exited after 0.05818 seconds with return value 0
Press any key to continue . . .
```

Question 28. Recursive function with a static variable

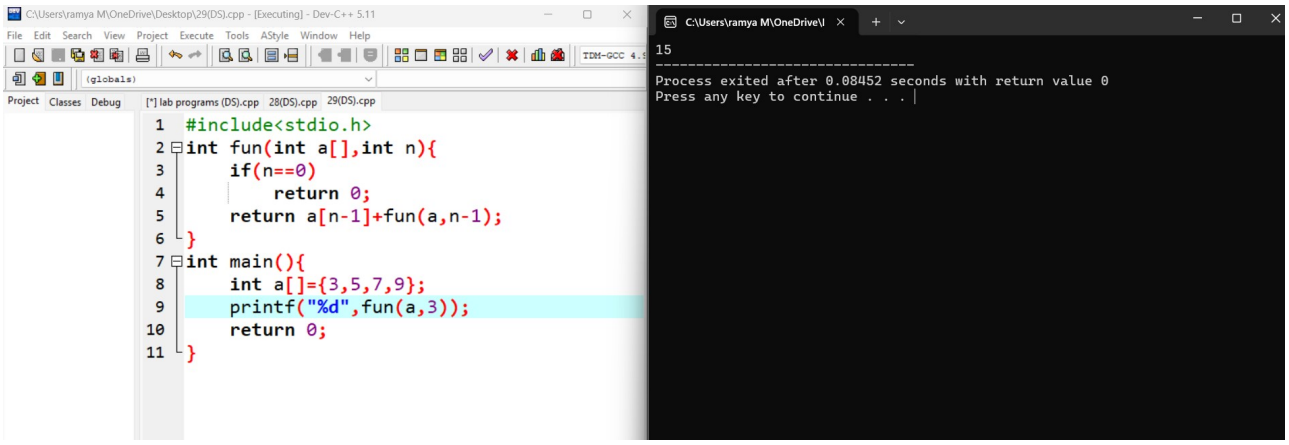


```
#include<stdio.h>
void fun(int n){
    static int x=1;
    if(n==0)
        return;
    x+=n;
    printf("%d",x);
    fun(n-1);
    printf("%d",x);
}
int main(){
    fun(3);
    return 0;
}
```

```
467777
--------------------------------
Process exited after 0.07633 seconds with return value 0
Press any key to continue . . .
```

--Reason: A static variable is shared across all recursive calls and is not restored to its previous value when a call returns.

Question 29. Recursive array processing



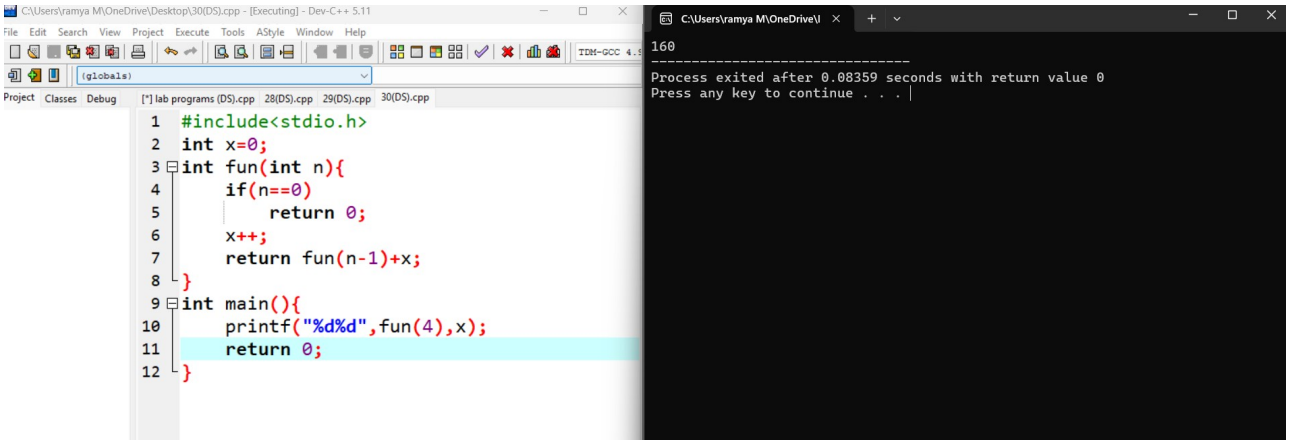
```
#include<stdio.h>
int fun(int a[],int n){
    if(n==0)
        return 0;
    return a[n-1]+fun(a,n-1);
}
int main(){
    int a[]={3,5,7,9};
    printf("%d",fun(a,3));
    return 0;
}
```

```
15
--------------------------------
Process exited after 0.08452 seconds with return value 0
Press any key to continue . . .
```

---The function adds the first three array elements:

The element 9 is not included because the function is called with n = 3.

Question 30. Recursion with a global variable



---the output is unspecified, either 16 4 or 16 0; the first value is always 16, but the second depends on the unspecified argument evaluation order, so no output is guaranteed.